

DMIMS Developer Blueprint

A. Product Overview

A1. Product Name

Datamation Inventory Management System

Short name:

DMIMS

Public URL:

<https://dmims.datamationgroup.com>

System type:

Datamation-hosted on-premise web platform

Technology:

Laravel + Filament + MySQL/MariaDB

A2. Purpose

DMIMS is a Datamation-hosted web platform for managing:

1. Customer companies
2. Customer users
3. Stock inventory
4. Document files
5. Archive boxes
6. Storage locations
7. Barcode scanning
8. Barcode printing
9. Internal and external movements
10. Subscriptions
11. Licenses
12. Manual billing and payment status

- 13. Reports and analytics
- 14. Audit logs
- 15. PWA/mobile-friendly access

All users log in from:

<https://dmims.datamationgroup.com/login>

Each customer company only sees its own data.

Datamation Super Admin can manage the full platform.

B. Core Development Rule

Every customer-owned table must include:

`customer_id`

Company users must only see:

records where `customer_id` = logged-in user's `customer_id`

Datamation Super Admin can see all records.

Never trust `customer_id` from browser request. Always use logged-in user context.

C. Final Role Model

To keep the system simple, Datamation only needs two internal roles.

C1. Datamation Super Admin

Full-control internal Datamation role.

Can manage:

Whole system

All customers

All users

Roles and permissions

Modules

Subscriptions

Licenses

Billing

Manual payment statuses

Invoices

Reports

Analytics

Audit logs

Backup / restore

System settings

Database rule:

users.customer_id = NULL

users.is_platform_user = true

role = Datamation Super Admin

C2. Datamation Management / Reports & Analytics

Read-only internal Datamation role.

Can view:

Platform dashboard

Customer summary

Subscription summary

License summary

Billing summary

Payment summary

Reports

Analytics

Audit summary if allowed

Cannot:

Create customers

Edit customers

Create users

Change subscriptions

Change licenses

Update billing

Update payment status

Delete records

Perform stock movements

Perform document movements

Change system settings

Database rule:

users.customer_id = NULL

users.is_platform_user = true

role = Datamation Management

C3. Customer Company Roles

Company Admin

Company Supervisor

Stock Inventory User

Document Tracking User

Viewer

D. System Architecture

DMIMS Platform

- |— Datamation Super Admin
- |— Datamation Management Reports & Analytics
- |— Customer Company Management
- |— User Management
- |— Roles & Permissions
- |— Module Control
- |— Subscription Control
- |— License Control
- |— Billing Management
- |— Manual Payment Status Management
- |— Shared Location Management
- |— Stock Inventory
- |— Document Tracking
- |— Movement Architecture

- └─ Barcode Registry
 - └─ Barcode Scanner
 - └─ Barcode Printing
 - └─ Reports
 - └─ Analytics
 - └─ Import / Export
 - └─ Audit Logs
 - └─ Notifications
 - └─ Backup / Restore
 - └─ System Settings
 - └─ PWA App Shell
-

E. Technology Stack

Layer	Technology
Backend	Laravel
Admin UI	Filament
Database	MySQL 8 / MariaDB
Authentication	Laravel / Filament Auth
Permissions	Spatie Laravel Permission

Frontend	Filament + Blade
Barcode	Code128 first, QR later
Scanner	USB barcode scanner first, browser camera later
Reports	CSV, Excel, PDF, Print View
Queue	Laravel database queue
Cache	File cache first, Redis later
PWA	Manifest + Service Worker
Server	Ubuntu Server
Web Server	Nginx
PHP	PHP-FPM
SSL	Cloudflare / Let's Encrypt
Backup	Automated database and file backup

F. Theme and UI Standard

F1. Theme Style

Use a clean Datamation-style Filament theme.

Area	Requirement
Sidebar	Dark navy
Content background	Light grey
Cards	White rounded cards
Primary buttons	Blue
Dangerous buttons	Red
Statuses	Colored badges
Tables	Search, filter, sort, pagination
Scanner pages	Large input and large buttons
Mobile	Touch-friendly and responsive

F2. Colors

Primary Navy: #0B1F3A

Primary Blue: #2563EB

Light Background: #F5F7FA

White Card: #FFFFFF

Success: #16A34A

Warning: #F59E0B

Danger: #DC2626

Text Dark: #111827

Muted Text: #6B7280

G. PWA Requirements

DMIMS must be PWA-ready.

Required:

manifest.json

service-worker.js

offline page

PWA icons

install prompt

responsive layout

mobile scanner page

HTTPS

standalone display mode

G1. Offline Behaviour

Login	Online required
Dashboard	Cached shell only
Product List	Cached shell, data online
Document List	Cached shell, data online
Scanner	Online required in version 1
Stock Movement	Online required
Document Movement	Online required
Reports	Online required

Offline message:

You are offline. Please reconnect to continue using DMIMS.

H. Main Navigation

H1. Datamation Super Admin Menu

Dashboard

Customers

Users

Roles & Permissions

Modules

Subscription Plans

Customer Subscriptions

Licenses

Billing

Payments

Invoices

Reports & Analytics

Audit Logs

Backup / Restore

System Settings

H2. Datamation Management Menu

Dashboard

Customer Summary

Subscription Summary

License Summary

Billing Summary

Payment Summary

Reports & Analytics

Audit Summary

Read-only only.

H3. Customer Company Menu

Dashboard

Stock Inventory

- └─ Categories
- └─ Products
- └─ Locations
- └─ Stock Receive-In
- └─ Stock Internal Transfer
- └─ Stock Out
- └─ Stock Adjustment
- └─ Stock Movements
- └─ Stock Reports

Document Tracking

- └─ Locations
- └─ Boxes
- └─ Document Files
- └─ File Receive-In
- └─ File Internal Transfer
- └─ File Move-Out / External Transfer
- └─ File Return
- └─ Box Receive-In
- └─ Box Internal Transfer
- └─ Box Move-Out / External Transfer
- └─ Box Return

- └─ Movement Logs
- └─ Document Reports

Shared Tools

- └─ Barcode Scanner
- └─ Barcode Registry
- └─ Barcode Printing
- └─ Reports
- └─ Audit Logs

Important:

Locations appear under both Stock Inventory and Document Tracking.

Both must use the same shared locations table.

Do not create stock_locations or document_locations.

I. Company Management

I1. Purpose

Used by Datamation Super Admin to manage customer companies.

I2. Company Statuses

trial

active

near_expiry

expired

suspended

cancelled

archived

I3. Functions

Create customer company

Edit customer company

Suspend customer company

Reactivate customer company

Archive customer company

View customer users

View enabled modules

View subscription

View license

View billing

View payment history

View reports

View audit logs

I4. Required Fields

company_name

company_code

status

Optional:

registration_no

tin_no

contact_person

email

phone

address

notes

J. User Management

J1. Functions

Create user

Edit user

Deactivate user

Assign company

Assign role

Reset password

View login history

Lock user after failed login attempts

J2. User Statuses

pending

active

inactive

suspended

locked

archived

J3. Rules

Email must be unique.

Company user must have customer_id.

Datamation internal user must have customer_id = NULL.

Company Admin can only manage own company users.

Company users cannot create platform users.

Suspended users cannot log in.

K. Module Control

K1. Modules

stock_inventory

document_tracking

barcode_scanning

barcode_printing

reports

import_export

advanced_audit

backup_restore

billing_view

K2. Rules

If module is disabled:

Hide menu.

Block route access.

Block direct URL access.

Block service/action execution.

Show message.

Message:

This module is not enabled for your company. Please contact Datamation.

L. Subscription Control

L1. Purpose

Subscription Control is the business/commercial entitlement layer.

Subscription answers:

What plan did the customer subscribe to?

Which modules are included?

What usage limits apply?

What is the validity period?

What is the billing cycle?

Subscription controls:

Plan

Modules

Limits

Valid dates

Billing cycle

Subscription does not decide final technical access alone.
Final access is confirmed by License Control.

L2. Subscription Controls These Items

Subscription plan

Enabled modules

Max users

Max products

Max document files

Max boxes

Allowed reports

Valid from date

Valid to date

Grace period

Billing cycle

Renewal period

Subscription status

L3. Subscription Plans

Example plans:

Plan	Users	Products	Files	Boxes	Modules
Trial	2	100	100	20	Stock only
Basic	5	1,000	5,000	500	Stock
Standard	20	10,000	50,000	5,000	Stock + Document
Enterprise	Custom	Custom	Custom	Custom	All

L4. Subscription Billing Cycles

monthly

quarterly

half_yearly

yearly

custom

Billing cycle is used to generate renewal reminders and billing records.

L5. Subscription Statuses

trial

active

near_expiry

expired_grace

expired

cancelled

Important:

Subscription suspended/revoked should not be used here.

Suspension and revocation belong to License Control.

L6. Subscription Expiry Logic

valid_to >= today

→ active

valid_to within next 30 days

→ near_expiry

valid_to < today but within grace_period_days

→ expired_grace

valid_to < today and grace period passed

→ expired

L7. Subscription Limit Checks

Before creating records, system must check subscription limits:

max_users

max_products

max_document_files

max_boxes

If subscription allows 20 users and license allows 15 users, use the stricter value:

effective max users = 15

Same rule applies to products, files and boxes.

Message:

Limit reached for your current subscription/license. Please contact Datamation.

L8. Subscription Renewal Flow

1. Datamation Super Admin opens Customer Subscription.
2. Click Renew Subscription.
3. Select same plan or new plan.
4. Set new valid_from and valid_to.
5. Set billing cycle.
6. Confirm modules and limits.
7. Save.
8. System updates customer_subscriptions.
9. System updates customer_modules.
10. System creates subscription log.
11. System creates audit log.
12. Super Admin may create billing record manually or automatically from subscription.

L9. SubscriptionService

Create:

app/Services/SubscriptionService.php

Required methods:

getCurrentSubscription(\$customerId)

getEnabledModules(\$customerId)

getSubscriptionLimits(\$customerId)

getBillingCycle(\$customerId)

daysUntilExpiry(\$customerId)

isSubscriptionActive(\$customerId)

isSubscriptionExpired(\$customerId)

isInGracePeriod(\$customerId)

checkUserLimit(\$customerId)

checkProductLimit(\$customerId)

checkDocumentFileLimit(\$customerId)

checkBoxLimit(\$customerId)

renewSubscription(\$customerId, \$data)

updateSubscriptionModules(\$customerId, \$modules)

M. License Control

M1. Purpose

License Control is the final technical access layer.

License answers:

Is this customer technically allowed to use the system now?

Should the system allow access, suspend access, or revoke access?

License controls:

active access

suspended access

revoked access

final technical permission to use the system

M2. License Does Not Replace Subscription

Subscription defines:

Plan

Modules

Limits

Valid dates

Billing cycle

License defines:

Whether the system should technically allow usage now

A customer may have a valid subscription but still be suspended by license.

Example:

Subscription = active until 31 Dec 2026

License = suspended due to payment/legal/admin issue

Result = operational access blocked

M3. License Statuses

active

suspended

revoked

expired

Use only these statuses to avoid confusion.

M4. License Behaviour

License Status	Behaviour
active	Customer can use system based on subscription, modules and permissions
suspended	Customer can login but only view/export, no operations
revoked	Customer access is blocked or Super Admin decides view-only
expired	Customer can login view/export only unless Super Admin renews

Recommended:

Suspended = login allowed, view/export only

Revoked = block login or Super Admin-controlled emergency access

Expired = view/export only

M5. License Functions

Datamation Super Admin can:

Create license

Assign license to customer

Activate license

Suspend license

Revoke license

Renew license

Set license valid from

Set license valid to

Set remarks

View license history

License should not duplicate plan/module configuration unless needed for stricter technical override.

Recommended license fields:

customer_id

license_no

license_status

valid_from

valid_to

grace_period_days

technical_access_mode

remarks

created_by

updated_by

M6. Technical Access Modes

full_access

view_only

blocked

Mapping:

active → full_access

suspended → view_only

expired → view_only

revoked → blocked

Super Admin may override manually if needed.

M7. License Renewal Flow

1. Datamation Super Admin opens Customer License.
2. Click Renew License.
3. Set new valid_from and valid_to.

4. Set license_status = active.
 5. Set technical_access_mode = full_access.
 6. Save.
 7. System writes license_logs.
 8. System writes audit_logs.
-

M8. License Suspension Flow

1. Datamation Super Admin opens Customer License.
 2. Click Suspend.
 3. Enter reason.
 4. System sets license_status = suspended.
 5. System sets technical_access_mode = view_only.
 6. System writes license_logs.
 7. System writes audit_logs.
-

M9. License Revocation Flow

1. Datamation Super Admin opens Customer License.
2. Click Revoke.
3. Enter reason.
4. System sets license_status = revoked.
5. System sets technical_access_mode = blocked.
6. System writes license_logs.
7. System writes audit_logs.

M10. LicenseService

Create:

app/Services/LicenseService.php

Required methods:

getCurrentLicense(\$customerId)

isLicenseActive(\$customerId)

isLicenseSuspended(\$customerId)

isLicenseRevoked(\$customerId)

isLicenseExpired(\$customerId)

getTechnicalAccessMode(\$customerId)

canUseSystem(\$customerId)

canPerformOperationalAction(\$customerId)

renewLicense(\$customerId, \$data)

suspendLicense(\$customerId, \$reason)

revokeLicense(\$customerId, \$reason)

N. Effective Access Rule

This is the most important combined rule.

A company user can perform operational actions only if:

Company status is active

AND user status is active

AND subscription is active or allowed grace period

AND license status is active

AND license technical_access_mode = full_access

AND module is enabled

AND user has permission

AND usage limit is not exceeded

A company user can view/export data if:

User is active

AND company is not archived/cancelled

AND license technical_access_mode is full_access or view_only

AND subscription is active, grace, expired or restricted-equivalent

AND user has view/export permission

Operational actions include:

Create

Edit

Delete

Stock receive-in

Stock out

Stock transfer

Stock adjustment

File receive-in

File transfer

File move-out

File return

Box receive-in

Box transfer

Box move-out

Box return

Import

Barcode registration

Barcode printing

O. Billing and Manual Payment Management

O1. Important Rule

There is no payment portal in this version.

Payment status is updated manually by Datamation Super Admin.

No online payment gateway is required.

O2. Purpose

Billing Management tracks:

Customer billing

Invoice records

Payment status

Outstanding amount

Subscription renewal billing

Manual payment confirmation

Billing remarks

O3. Billing Functions

Datamation Super Admin can:

Create billing record

Create invoice record

Assign billing to customer

Link billing to subscription

Link billing to license

Set billing period

Set invoice number

Set invoice date

Set due date

Set amount

Set tax amount

Set total amount

Set payment status manually

Add payment remarks

Upload payment proof if needed

Mark as paid

Mark as unpaid

Mark as partially paid

Mark as overdue

Cancel billing record

View billing history

Export billing report

O4. Payment Statuses

unpaid

paid

partially_paid

overdue

waived

cancelled

refunded

O5. Billing Statuses

draft

issued

sent

partially_paid

paid

overdue

cancelled

void

O6. Manual Payment Update Flow

1. Datamation Super Admin opens Billing.
2. Select customer billing record.
3. Click Update Payment Status.
4. Select payment status.
5. Enter paid amount.
6. Enter payment date.
7. Enter payment method.
8. Enter reference number.
9. Upload proof if available.
10. Add remarks.
11. Save.
12. System updates billing record.
13. System writes payment record.
14. System writes audit log.
15. If payment confirms renewal, Super Admin renews subscription/license manually.

Billing does not automatically renew subscription/license unless Datamation later chooses to automate it.

O7. Payment Methods

Bank Transfer

Cash

Cheque

Online Transfer

Internal Adjustment

Waived

Other

O8. Billing Rules

- Billing does not automatically activate subscription.
 - Super Admin manually confirms renewal.
 - Payment status must not be editable by customer users.
 - Management users can only view billing summaries.
 - Company users may view own billing status only if billing_view module is enabled.
 - All billing/payment changes must be audited.
-

P. Movement Architecture

P1. Core Philosophy

DMIMS supports three movement types.

Movement Type	Meaning
Internal Transfer	DMIMS → DMIMS
External Receive-In	External Source → DMIMS
External Move-Out	DMIMS → External Destination

Do not mix them.

Do not create fake locations like:

CUSTOMER

LAWYER

OUTSIDE

UNKNOWN

TEMP STORAGE

External sources and destinations must be stored in structured movement fields.

P2. Internal Transfer

Internal Transfer means movement between records already inside DMIMS.

Examples:

Product from Warehouse A to Warehouse B

File from Box A to Box B

Box from Archive Room to Legal Room

Rules:

Source must exist in DMIMS.

Destination must exist in DMIMS.

Same customer_id required.

Movement log required.

Audit log required.

P3. External Receive-In

External Receive-In means item enters DMIMS from an external source.

Examples:

Supplier delivers products

Customer sends files

Lawyer returns documents

External branch sends boxes

Manual discovered stock

Old archive onboarding

Rules:

External source does not need to exist in DMIMS.

Use source_type, source_name, source_reference_no.

Destination must exist in DMIMS.

Movement log required.

Audit log required.

P4. External Move-Out

External Move-Out means item leaves DMIMS to an external destination.

Examples:

File sent to lawyer

Box sent to archive vendor

Product sent outside company

File handed to staff

Rules:

External destination does not need to exist in DMIMS.

Use destination_type, destination_name, destination_reference_no.

Preserve last known internal box/location.

Movement log required.

Audit log required.

Return flow required if temporary.

Q. Stock Inventory

Q1. Pages

Categories

Products

Locations

Stock Receive-In

Stock Internal Transfer

Stock Out

Stock Adjustment

Stock Movements

Stock Reports

Q2. Stock Receive-In

Required fields:

Product

Destination location

Quantity

Source type

Source name

Source reference number

Remarks

Performed by

Receive datetime

System updates:

Increase stock quantity

Create stock_movements record

movement_type = receive_in

Write audit log

Q3. Stock Internal Transfer

Required fields:

Product

Source location

Destination location

Quantity

Transfer reason

Remarks

Performed by

Transfer datetime

System updates:

Deduct source quantity

Add destination quantity

movement_type = internal_transfer

Write audit log

Q4. Stock Out

Required fields:

Product

Source location

Quantity

Destination type

Destination name

Destination reference number

Reason

Remarks

Performed by

Stock out datetime

System updates:

Decrease stock quantity

movement_type = stock_out

Write audit log

Q5. Stock Adjustment

Required fields:

Product

Location

Adjustment quantity

Reason

Remarks

Performed by

Adjustment datetime

Rules:

Reason required.

Movement log required.

Audit log required.

R. Document Tracking

R1. Pages

Locations

Boxes

Document Files

File Receive-In

File Internal Transfer

File Move-Out / External Transfer

File Return

Box Receive-In

Box Internal Transfer

Box Move-Out / External Transfer

Box Return

Movement Logs

Document Reports

R2. File Receive-In

Required fields:

File barcode or auto-generate

File reference number

Document type

Destination box

Source type

Source name

External contact person

Source reference number

Receive datetime

Remarks

Received by

System updates:

Create document file

Register barcode

action_type = receive_in_file

Write audit log

R3. Box Receive-In

Required fields:

Box number

Destination location

Source type

Source name

Source reference number

Receive datetime

Remarks

Received by

System updates:

Create box

Assign location

action_type = receive_in_box

Write audit log

R4. File Internal Transfer

Required fields:

File barcode

Current box

Destination box

Transfer reason

Remarks

Performed by

Transfer datetime

System updates:

Update document_files.current_box_id

Update box file counts

action_type = transfer_file

Write audit log

R5. Box Internal Transfer

Required fields:

Box barcode

Current location

Destination location

Transfer reason

Remarks

Performed by

Transfer datetime

System updates:

Update boxes.current_location_id

action_type = transfer_box

Write audit log

Important:

Do not update every file location.

File location = Document File → Current Box → Current Location

R6. File Move-Out

Required fields:

File barcode

Destination type

Destination name

Contact person

Contact phone/email

External address

Expected return date

Handover reference number

Reason

Remarks

Performed by

Transfer datetime

System updates:

`document_files.current_status = moved_out`

`document_files.last_internal_box_id = previous current_box_id`

Keep `current_box_id` as last known box

action_type = move_out_file

Write audit log

R7. Box Move-Out

Required fields:

Box barcode

Destination type

Destination name

Contact person

Contact phone/email

External address

Expected return date

Handover reference number

Reason

Remarks

Performed by

Transfer datetime

System updates:

boxes.status = moved_out

boxes.last_internal_location_id = previous current_location_id

Keep current_location_id as last known location

action_type = move_out_box

Write audit log

R8. File Return

Required fields:

File barcode

Return box

Return remarks

Returned by

Received by

Return datetime

System updates:

`document_files.current_status = active`

`document_files.current_box_id = return box`

`document_files.actual_return_date = return datetime`

`action_type = return_file`

Write audit log

R9. Box Return

Required fields:

Box barcode

Return location

Return remarks

Returned by

Received by

Return datetime

System updates:

boxes.status = active

boxes.current_location_id = return location

boxes.actual_return_date = return datetime

action_type = return_box

Write audit log

S. Shared Location Management

Locations are physical places inside DMIMS control.

Rules:

Locations are shared.

Products are stored in locations.

Boxes are stored in locations.

Files are stored inside boxes.

Boxes are not locations.

Do not create fake external locations.

T. Barcode System

T1. Barcode Types

Product

Location

Box

Document File

T2. Barcode Format

PRD-COMPANYCODE-000001

LOC-COMPANYCODE-000001

BOX-COMPANYCODE-000001

DOC-COMPANYCODE-000001

T3. Scan Flow

1. User scans barcode
2. Check barcode_registry
3. Detect barcode type
4. Check customer_id
5. Check permission
6. Open correct page/action
7. Write barcode_scan_logs

If barcode belongs to another company:

Barcode not found or access denied.

U. Dashboard, Reports and Analytics

U1. Datamation Super Admin Dashboard

Total customers

Active customers

Suspended customers

Near expiry subscriptions

Expired subscriptions

Active licenses

Suspended licenses

Revoked licenses

Unpaid invoices

Overdue billing

Total revenue recorded manually

Recent audit activities

U2. Datamation Management Dashboard

Read-only analytics:

Customer count

Subscription trend

License status trend

Billing summary

Payment summary

Outstanding balance

Monthly billing amount

Active vs suspended customers

Module usage summary

U3. Customer Dashboard

Total products

Low stock items

Total boxes

Total document files

Recent stock receive-ins

Recent stock outs

Files moved out

Boxes moved out

Overdue file returns

Overdue box returns

Subscription status

License status

V. Database Structure

V1. Core Tables

customers

users

roles

permissions

model_has_roles

model_has_permissions

modules

customer_modules
subscription_plans
customer_subscriptions
subscription_logs
licenses
license_logs
billing_records
billing_payments
billing_logs
settings
audit_logs
notifications

V2. Stock Tables

categories
products
product_location_stocks
stock_movements
stock_alerts

V3. Document / Location / Barcode Tables

locations
location_types
boxes
document_types

document_files

document_movement_logs

barcode_registry

barcode_scan_logs

W. Required Models, Services, Middleware and Resources

W1. Models

Customer

User

Module

CustomerModule

SubscriptionPlan

CustomerSubscription

SubscriptionLog

License

LicenseLog

BillingRecord

BillingPayment

BillingLog

Setting

Category

Product

Location

LocationType

ProductLocationStock

StockMovement

StockAlert

Box

DocumentType

DocumentFile

DocumentMovementLog

BarcodeRegistry

BarcodeScanLog

AuditLog

Notification

W2. Services

CompanyContextService

UserSecurityService

ModuleAccessService

SubscriptionService

LicenseService

BillingService

PaymentService

BarcodeService

ScannerService

LocationService

StockMovementService

DocumentMovementService

AuditService

NotificationService

ReportExportService

ImportService

BackupService

AccessControlService

W3. AccessControlService

Create one combined access service so developers do not duplicate checks everywhere.

canLogin(\$user)

canView(\$user, \$module)

canExport(\$user, \$module)

canPerformOperationalAction(\$user, \$module, \$action)

getEffectiveLimits(\$customerId)

getEffectiveAccessMode(\$customerId)

This service should combine:

company status

user status

subscription status

license status

module access

permissions

limits

W4. Middleware

EnsureUserIsActive

EnsureCompanyAssigned

EnsureCompanyActive

EnsureSubscriptionValid

EnsureLicenseAllowsAccess

EnsureModuleEnabled

SetCompanyContext

LogUserActivity

X. Security, Audit and Performance

X1. Security Rules

Never trust customer_id from browser.

Always use logged-in user context.

Use middleware, policies, permissions and services.

Block direct URL access.

APP_DEBUG=false in production.

HTTPS required.

.env must not be exposed.

Billing/payment edits only by Datamation Super Admin.

Management role is read-only.

X2. Audit These Actions

Login

Logout

Failed login

User created/updated/deactivated

Role changed

Permission changed

Customer created/suspended

Subscription created/renewed/expired/cancelled

License created/renewed/suspended/revoked

Billing created/updated/cancelled

Payment status updated

Module enabled/disabled

Product created/updated

Stock receive-in/out/internal transfer/adjustment

Box receive-in/internal transfer/move-out/return

File receive-in/internal transfer/move-out/return

Barcode scanned/printed

Import performed

Export performed

Backup performed

Restore performed

Y. Development Phases

1. Run existing source code
 2. Add customers
 3. Add customer_id isolation
 4. Add simplified Datamation roles
 5. Add company users
 6. Add roles and permissions
 7. Add module control
 8. Add subscription control
 9. Add license control
 10. Add AccessControlService
 11. Add manual billing/payment management
 12. Add shared locations
 13. Add movement architecture
 14. Complete stock module
 15. Complete document tracking module
 16. Add barcode registry
 17. Add scanner and printing
 18. Add reports and analytics
 19. Add audit logs
 20. Add import/export
 21. Add backup/restore
 22. Add PWA support
 23. Prepare production deployment
-

Z. Final Acceptance Checklist

Platform:

Application installs cleanly

Datamation Super Admin can login

Datamation Management user is read-only

Customer can be created

Company users can be created

Company isolation works

Roles and permissions work

Modules can be enabled/disabled

PWA works

Audit logs work

Reports export correctly

Subscription:

Subscription controls plan, modules, limits, dates and billing cycle

Subscription plan can be created

Customer subscription can be assigned

Renewal works

Grace period works

Max user/product/file/box limits work

Disabled module is hidden and blocked

Subscription logs are created

License:

License controls final technical access

License can be created

License can be assigned to customer

License active allows access

License suspended allows view/export only

License revoked blocks access

License expired allows view/export only

License logs are created

Billing:

Billing record can be created

Invoice number can be stored

Payment status can be manually updated by Super Admin

Partial payment works

Overdue billing appears

Paid amount and balance are calculated

Payment logs are created

Management role can view summaries only

Customer users cannot edit billing/payment

Access Control:

Operational action only works when company, user, subscription, license, module and permission are valid

View/export works in view-only mode

Direct URL access is blocked

Stricter limit between subscription and license is enforced if license limit exists

Movement:

Internal transfer requires DMIMS source and destination

External receive-in allows external source details

External move-out allows external destination details

No fake external locations are needed

File return works

Box return works

Box movement does not update every file

Overdue return reports work

External movement reports work

Security:

APP_DEBUG=false

HTTPS enabled

.env not exposed

Company users cannot see other company data

Management users are read-only

All mutation actions are audited

Final Developer Instruction

Use this practical control model:

Subscription controls:

- plan

- modules

- limits

- valid dates
- billing cycle

License controls:

- active access
- suspended access
- revoked access
- final technical permission to use the system

Billing controls:

- invoice/billing record
- manual payment status
- paid/unpaid/partial/overdue tracking

The first major milestone is:

Datamation Super Admin can create Company A and Company B.

Company A user sees only Company A data.

Company B user sees only Company B data.

Subscription controls plan/modules/limits/dates.

License controls final access mode.

Billing/payment is manually updated by Super Admin.

Management user can only view reports and analytics.

Movement architecture separates Internal Transfer, External Receive-In and External Move-Out.

Locations remain clean and are not polluted with fake external destinations.

Build step by step. Do not build everything at once.